



UNIVERSIDAD SERGIO ARBOLEDA

Sistemas Operativos

Taller #02

Taller protocolo SSH

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1. Prueba de conexión

Realizar una prueba de conectividad con el comando *ping* hacia la otra máquina virtual, para comprobar que ambas máquinas sean capaces de comunicarse entre sí a través de esa red.

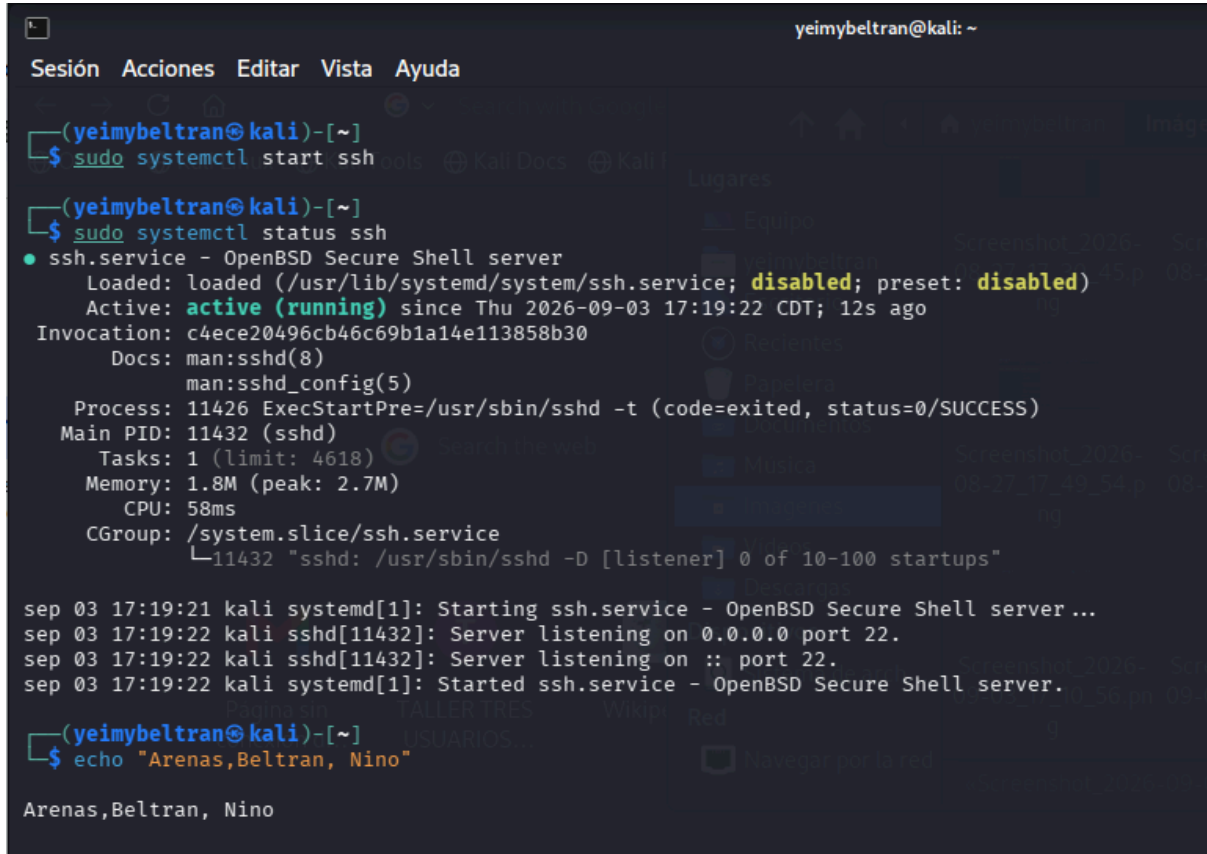
```
(yeimbeltran@kali)-[~]
$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data:
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=1.57 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=0.634 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=0.549 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=0.728 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=0.535 ms
64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=0.582 ms
64 bytes from 192.168.56.101: icmp_seq=7 ttl=64 time=0.604 ms
64 bytes from 192.168.56.101: icmp_seq=8 ttl=64 time=0.718 ms
64 bytes from 192.168.56.101: icmp_seq=9 ttl=64 time=0.751 ms
64 bytes from 192.168.56.101: icmp_seq=10 ttl=64 time=0.654 ms
64 bytes from 192.168.56.101: icmp_seq=11 ttl=64 time=0.672 ms
^C
— 192.168.56.101 ping statistics —
11 packets transmitted, 11 received, 0% packet loss, time 10191ms
rtt min/avg/max/mdev = 0.535/0.726/1.566/0.274 ms

(yeimbeltran@kali)-[~]
$

(yeimbeltran@kali)-[~]
$ echo "Arenas,Beltran, Nino"
```

2. Verificación de SSH

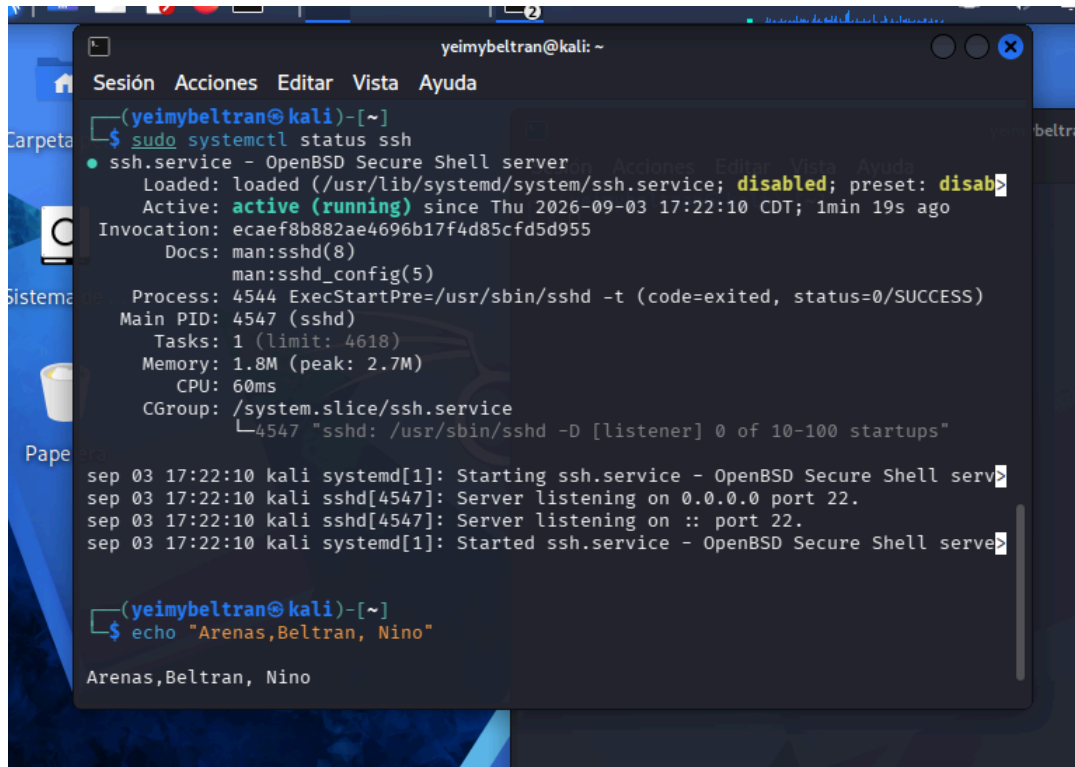
Verificar la instalación y disponibilidad del protocolo SSH en las máquinas virtuales, comprobando que el servicio estuviera habilitado para permitir conexiones remotas.



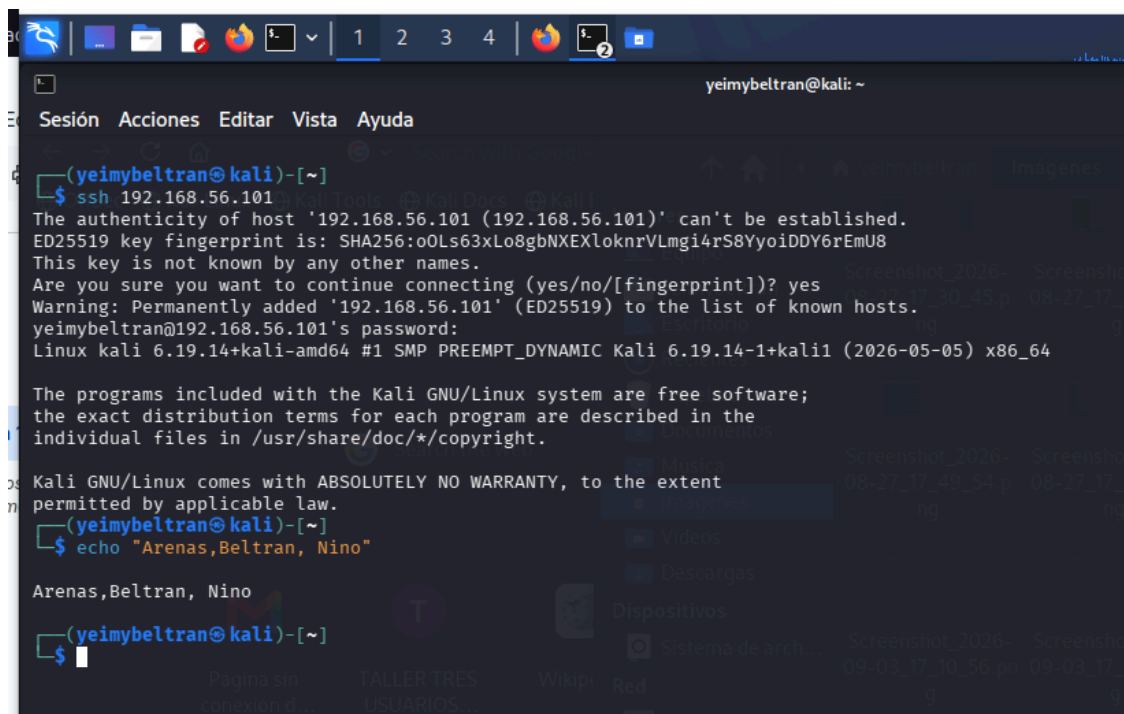
```
yeimybeltran@kali: ~  
Sesión Acciones Editar Vista Ayuda  
(yeimybeltran@kali)-[~]  
$ sudo systemctl start ssh  
(yeimybeltran@kali)-[~]  
$ sudo systemctl status ssh  
● ssh.service - OpenBSD Secure Shell server  
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)  
   Active: active (running) since Thu 2026-09-03 17:19:22 CDT; 12s ago  
 Invocation: c4ece20496cb46c69b1a14e113858b30  
    Docs: man:sshd(8)  
          man:sshd_config(5)  
  Process: 11426 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)  
 Main PID: 11432 (sshd)  
    Tasks: 1 (limit: 4618)  
  Memory: 1.8M (peak: 2.7M)  
     CPU: 58ms  
   CGroup: /system.slice/ssh.service  
           └─11432 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"  
  
sep 03 17:19:21 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server ...  
sep 03 17:19:22 kali sshd[11432]: Server listening on 0.0.0.0 port 22.  
sep 03 17:19:22 kali sshd[11432]: Server listening on :: port 22.  
sep 03 17:19:22 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.  
(yeimybeltran@kali)-[~]  
$ echo "Arenas,Beltran, Nino"  
Arenas,Beltran, Nino
```

3. Conexión mediante SSH

Se utiliza este protocolo para establecer la conexión remota con la otra máquina virtual. De esta manera, fue posible acceder a esta y ejecutar comandos de forma remota.



```
yeimibeltran@kali: ~  
Sesión Acciones Editar Vista Ayuda  
(yeimibeltran@kali)-[~]  
$ sudo systemctl status ssh  
● ssh.service - OpenBSD Secure Shell server  
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)  
   Active: active (running) since Thu 2026-09-03 17:22:10 CDT; 1min 19s ago  
 Invocation: ecae8b882ae4696b17f4d85cfd5d955  
    Docs: man:sshd(8)  
          man:sshd_config(5)  
   Process: 4544 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)  
   Main PID: 4547 (sshd)  
     Tasks: 1 (limit: 4618)  
    Memory: 1.8M (peak: 2.7M)  
       CPU: 60ms  
   CGroup: /system.slice/ssh.service  
           └─4547 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"  
  
sep 03 17:22:10 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell serv  
sep 03 17:22:10 kali sshd[4547]: Server listening on 0.0.0.0 port 22.  
sep 03 17:22:10 kali sshd[4547]: Server listening on :: port 22.  
sep 03 17:22:10 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell serve  
  
(yeimibeltran@kali)-[~]  
$ echo "Arenas,Beltran, Nino"  
Arenas,Beltran, Nino
```



```
yeimibeltran@kali: ~  
Sesión Acciones Editar Vista Ayuda  
(yeimibeltran@kali)-[~]  
$ ssh 192.168.56.101  
The authenticity of host '192.168.56.101 (192.168.56.101)' can't be established.  
ED25519 key fingerprint is: SHA256:oOLs63xLo8gbNXEXloknrVLmgi4rS8YyoiDDY6rEmU8  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.56.101' (ED25519) to the list of known hosts.  
yeimibeltran@192.168.56.101's password:  
Linux kali 6.19.14+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05) x86_64  
  
The programs included with the Kali GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
(yeimibeltran@kali)-[~]  
$ echo "Arenas,Beltran, Nino"  
Arenas,Beltran, Nino  
  
(yeimibeltran@kali)-[~]  
$
```

Una vez exista la conexión, se crea una carpeta y dentro de ella un archivo *.txt*, para luego comprobar que los cambios realizados mediante SSH sean visibles en la otra máquina.

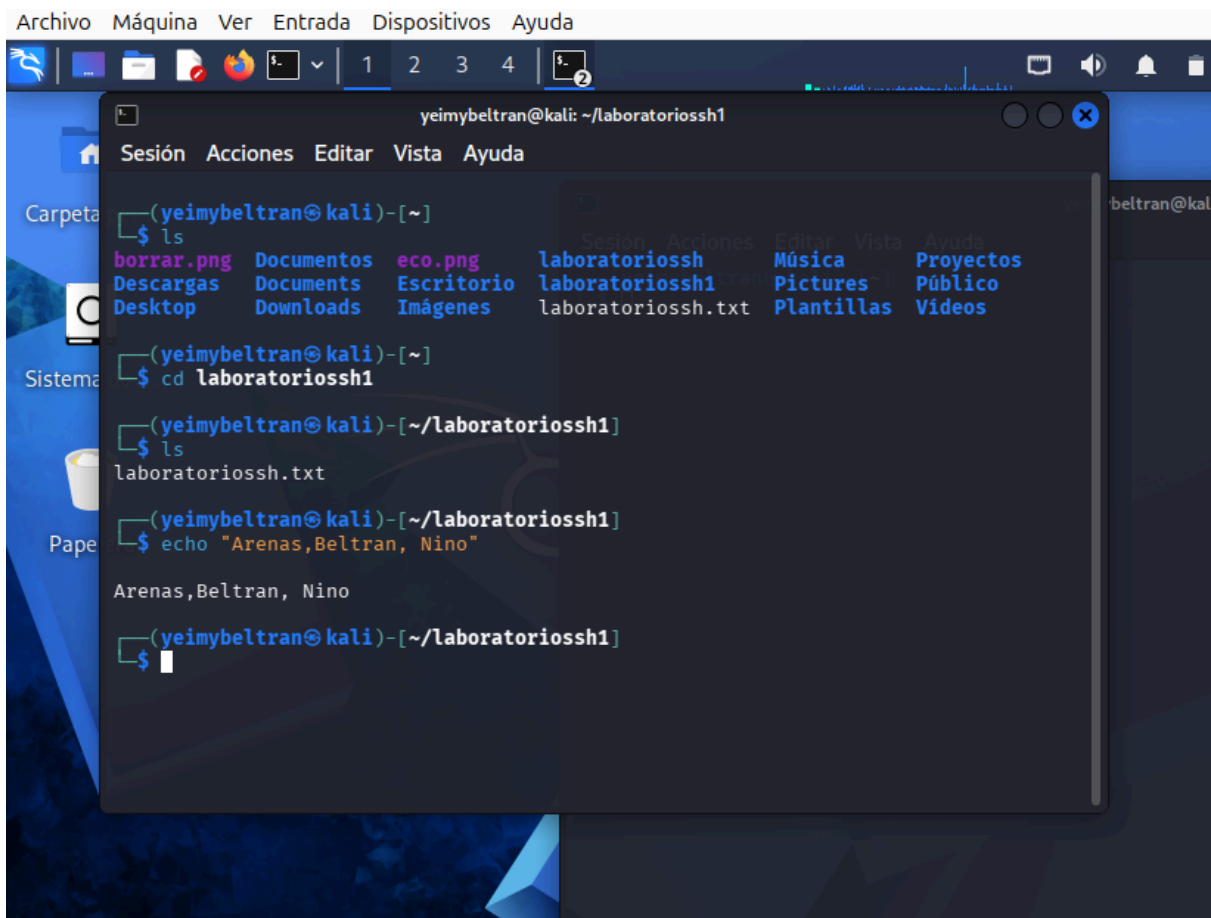
```
(yeimbeltran@kali)-[~]
$ cd laboratoriossh1

(yeimbeltran@kali)-[~/laboratoriossh1]
$ nano laboratoriossh.txt

(yeimbeltran@kali)-[~/laboratoriossh1]
$ echo "Arenas,Beltran, Nino"

Arenas,Beltran, Nino

(yeimbeltran@kali)-[~/laboratoriossh1]
$
```



4. Edición del archivo

Se edita el archivo .txt para agregar los nombres de los integrantes y verificar el contenido del archivo.

